

David Jones Marketing Campaign Strategy: Analytics and Database Design Report

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Part A – Database Design and Development

1.1 Entity Identification

Entity Name	Entity Description
CUSTOMERS	Represents individual customers at David Jones, storing their contact details, geographical location, and loyalty program status.
PRODUCTS	Stores details for all merchandise, including brand, category, pricing, and current inventory levels.
ORDERS	Stores details of transaction data, including order date and payment methods that customers used on the website.
ORDER_PRODUCTS	Details the specific items and quantities contained within each order.
CAMPAIGNS	Stores details for all campaigns, including the campaign names and objectives.
ADVERTISEMENTS	Stores details of an advertisement from David Jones, including the platform and device used.
INTERACTION_EVENTS	Stores how customers engage with each advertisement, including the duration of each interaction.
ADVERTISEMENT_PERFORMANCE	Tracks the performance of each advertisement, including the number of clicks, impressions, and conversions.
ADVERTISEMENT_ORDER	Represents whether David Jones acquire orders through advertisements or through direct purchases.
DISCOUNTS	Represents details of a discount event.
ORDER_PRODUCT_DISCOUNT	Represents the connection between discounts for product lines.
ORDER_DISCOUNT	Represents the discounts for the total order.

1.2 Identify Attributes, Primary Keys and Foreign Keys

Entity Name	Attributes
CUSTOMERS	8 attributes: customer_id (PK) , customer_name, customer_email, location, phone_number, referrer_id (FK) , signup_date, loyalty_tier
PRODUCTS	7 attributes: product_id (PK) , product_name, category, brand, unit_price, unit_cost, stock_quantity
ORDERS	5 attributes: order_id (PK) , customer_id (FK) , order_date, payment_method, order_status
ORDER_PRODUCTS	4 attributes: order_id (PK) , product_id (PK) , quantity, gross_price
CAMPAIGNS	6 attributes: campaign_id (PK) , campaign_name, objective, total_budget, start_date, end_date
ADVERTISEMENTS	6 attributes: ad_id (PK) , campaign_id (FK) , platform, allocated_budget, start_date, end_date
INTERACTION_EVENTS	6 attributes: interaction_id (PK) , ad_id (FK) , customer_id (FK) , duration_seconds, interaction_datetime, device
ADVERTISEMENT_PERFORMANCE	8 attributes: ad_performance_id (PK) , ad_id (FK) , reported_date, device, impressions, clicks, conversions, platform_ad_revenue
ADVERTISEMENT_ORDER	3 attributes: primary key (ad_id, order_id) , clicked_date
DISCOUNTS	10 attributes: discount_id (PK) , campaign_id (FK) , discount_name, discount_type, scope, discount_value, min_spend, discount_start_date, discount_end_date
ORDER_PRODUCT_DISCOUNT	4 attributes: primary key (order_id, product_id, discount_id) , product_discount_amount
ORDER_DISCOUNT	3 attributes: primary key (order_id, discount_id) , order_discount_amount

1.3 Relationship Definition and Cardinality Analysis

Customer places Order. This is a **1-Many relationship** where 1 customer can place many orders but 1 order can only be placed by 1 customer.

Order contains Product. This is a **M-M relationship**. 1 order can contain many products and 1 product can be in many orders, creating an **associative entity** called Order Product.

Customer interacts with Advertisement. This is a **M-M relationship**. 1 customer can interact with many ads and 1 ad can be interacted by many customers. This creates an **associative entity** called Interaction Event.

Advertisement results in Order. This is a **M-M relationship**. 1 ad can result in many orders and 1 order can come from many ads. This creates an **associative entity** called Advertisement Order.

Advertisement is part of Campaign. This is a **1-M relationship**. 1 campaign can have many ads but 1 ad can only belong to 1 campaign.

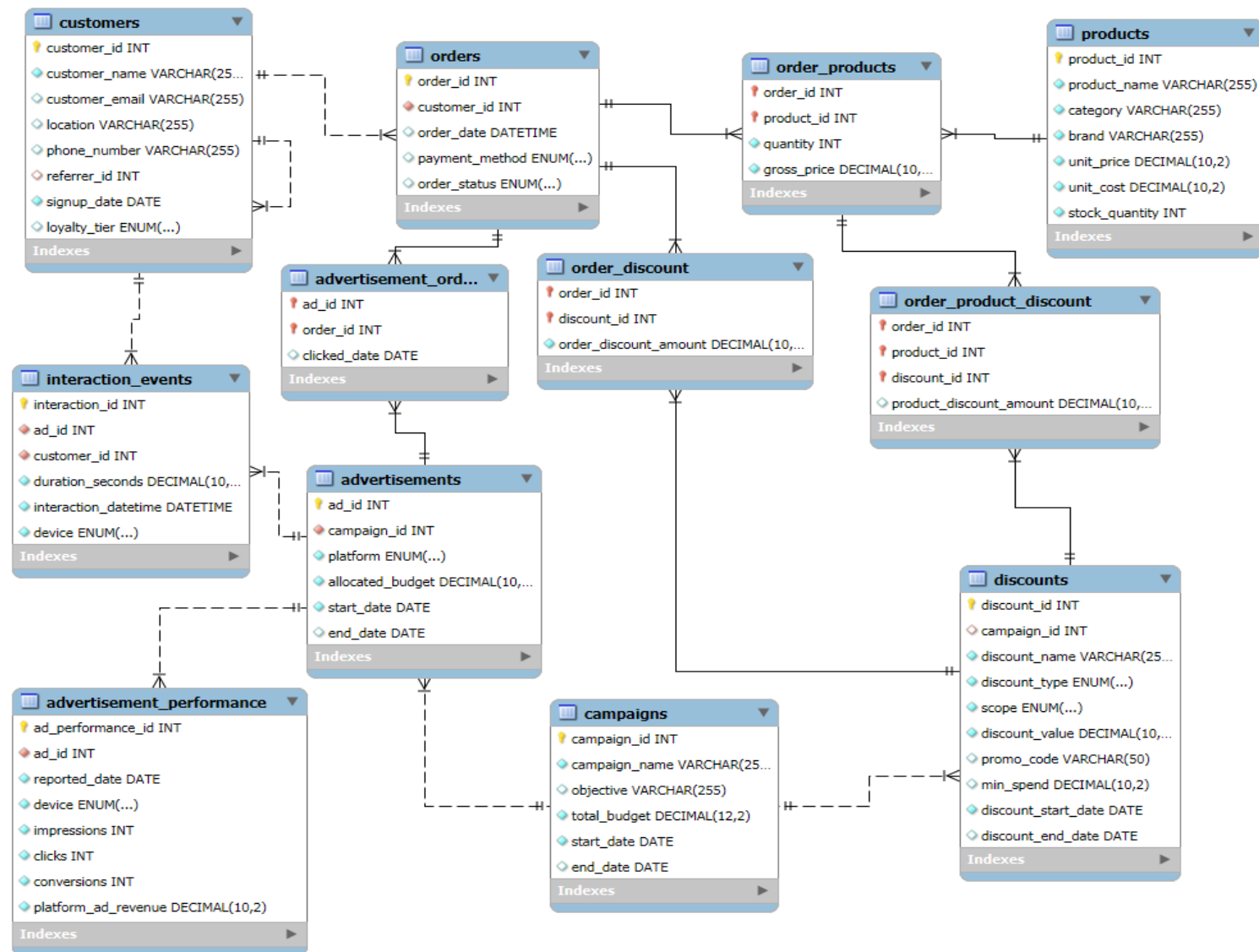
Advertisement tracks Advertisement Performance. This is a **1-M relationship**. 1 ad has many performances but 1 ad performance is tied to only 1 ad.

Discounts is part of Campaign. This is a **1_M relationship**. 1 campaign can have many discounts but 1 discount can only be traced back to 1 campaign.

Discounts are applied in orders. This is a **M-M relationship**. 1 discount can be applied to many orders and 1 order can use many discounts. This creates an **associative entity** called order_discount.

Order_product can have discounts. This is a **M-M relationship**. A discount applies to many order-lines, and a line could carry more than one discount. This creates an **associative entity** called order_product_discount.

1.4 ERD diagram



1.5 Assumptions and Dummy Data Limitations

This analysis is built on a synthetic dataset generated in Python and implemented in MySQL Workbench, created for academic purposes to demonstrate database design and SQL analytical skills. The data was modelled to reflect realistic online retail and marketing behavior for David Jones across the period 1 January 2021 to 31 May 2026. The following assumptions and limitations should frame how the findings are interpreted.

Recommendations are drawn from a simulated dataset and exclude external factors such as economic conditions and competitor activity. Before committing spend, the same analysis should be re-run on live campaign data to validate the patterns observed here.

- 1 **Membership and channel scope.** The dataset covers 400 customers, all assumed to be David Jones loyalty members across three tiers (Bronze, Silver and Gold), with every transaction occurring exclusively through the online store. In-store sales, non-member purchases and other channels are out of scope, so all results describe online member behavior only.
- 2 **Simulated behavior.** Records were generated to produce realistic but deliberately patterned behavior: repeat purchasing, tier-based spending differences, and a clear split between direct and advertisement-influenced orders. Because the data is synthetic, the observed relationships are properties of the generation logic rather than evidence of real-world customer behavior, and should not be read as benchmarks for the actual business.
- 3 **Advertisement attribution.** An order is linked to an advertisement through the advertisement_order bridge whenever a customer clicked that ad before purchasing. This is a simplified, click-based attribution model: it does not weight multiple touchpoints, account for view-through influence, or apply a time-decay window. Attributed orders and revenue should therefore be treated as indicative rather than precise.
- 4 **Two distinct measures of revenue.** The report intentionally separates platform-reported ad revenue (platform_ad_revenue, used for campaign ROI in Concern 2) from actual completed-order revenue (quantity $\times$ gross_price, used in Concerns 3, 7 and 9). These two figures are not expected to reconcile, wherein the former reflects platform-side estimates while the latter reflects realized transactional sales, so comparisons across concerns must respect this distinction.
- 5 **Order status and post-sale adjustments.** Revenue, profit and customer-value queries count only orders with a status of 'Completed', excluding pending, cancelled or returned orders. Returns and refunds after completion are not modelled, so all profit figures represent gross realised profit without post-sale adjustments.
- 6 **Discounts modelled but not applied.** The schema includes a complete discounting structure (discounts, order_discount and order_product_discount), but no discounts were populated during the analysis window. All revenue and profit figures therefore reflect undiscounted gross pricing, which will overstate net margins relative to a real EOFY sale period where markdowns are central to the campaign.
- 7 **Non-converting traffic.** One customer (customer_id = 13) was deliberately recorded as viewing advertisements without ever placing an order, enabling analysis of non-converting

ad traffic. This single case is illustrative only and does not represent a realistic non-converter rate.

- 8 **Simplified marketing metrics.** CTR (clicks / impressions), click-to-conversion (conversions / clicks), CPA (spend / conversions) and ROI ((revenue – spend) / spend) are computed directly from the recorded fields. They exclude external influences such as economic conditions, seasonality, competitor activity and ad fatigue, and assume that reported impressions, clicks and conversions are accurate and free of bot or duplicate activity.
- 9 **Illustrative reference data.** Campaign names, brands, product names and locations are illustrative and do not reflect actual David Jones strategy, merchandise or store performance. Customer demographics are limited to basic contact and location details, so segmentation by age, gender or income is not possible.

Part B – Business Analytics Report

1.6 Business concern 1

Name and Description: Platform CTR and Conversion rate. This query addresses a business concern from the CMO: which platform has the highest CTR and conversion rate (from advertisements to actual orders).

```
select
  a.platform,
  sum(ap.impressions) as total_impressions,
  sum(ap.clicks) as total_clicks,
  sum(ap.conversions) as total_conversions,
  round(sum(ap.clicks) / nullif(sum(ap.impressions), 0) * 100, 2) as ctr_pct,
  round(sum(ap.conversions) / nullif(sum(ap.clicks), 0) * 100, 2) as click_to_conversion_pct
from advertisements a
join advertisement_performance ap on a.ad_id = ap.ad_id
group by a.platform
order by ctr_pct desc;
```

	platform	total_impressions	total_clicks	total_conversions	ctr_pct	click_to_conversion_pct
►	Instagram	182384947	10970634	1647318	6.02	15.02
	Facebook	159531966	6355616	1172505	3.98	18.45
	WeChat	135616540	4156781	1038501	3.07	24.98
	X	133543854	2798015	256391	2.10	9.16

Justification. The CMO needs to know where advertising spend actually converts. CTR captures whether the creative earns the click (top of funnel), while the click-to-conversion rate captures whether that click becomes a paid order (mid funnel). Separating the two stops a platform that generates cheap, low-intent clicks from being mistaken for a strong performer.

Insight. Instagram leads decisively on CTR at 6.02%, ahead of Facebook (3.98%), WeChat (3.07%) and X (2.10%). The funnel then reverses: WeChat converts 24.98% of its clicks into orders, versus 18.45% for Facebook, 15.02% for Instagram and just 9.16% for X. Instagram is the strongest acquisition engine; WeChat is the strongest closer.

Addresses the concern? Yes. It answers the CMO directly, but shows the highest CTR and the highest conversion rate belong to two different platforms, so the recommendation is a channel pairing rather than a single winner.

1.7 Business concern 2

Name and Description: Campaign ROI. This query measures the Return-on-Investment (ROI) of each campaign to discover the most efficient and profitable marketing campaign.

```
with spend as (  
    select campaign_id, sum(allocated_budget) as total_ad_spend  
    from advertisements  
    group by campaign_id  
)  
  
revenue as (  
    select a.campaign_id, sum(ap.platform_ad_revenue) as total_ad_revenue  
    from advertisements a  
    join advertisement_performance ap on a.ad_id = ap.ad_id  
    group by a.campaign_id  
)  
  
select  
    c.campaign_name,  
    s.total_ad_spend,  
    coalesce(r.total_ad_revenue, 0) as total_ad_revenue,  
    round(((coalesce(r.total_ad_revenue, 0) - s.total_ad_spend) / nullif(s.total_ad_spend, 0) * 100, 2) as roi_pct  
from campaigns c  
join spend s on c.campaign_id = s.campaign_id  
left join revenue r on c.campaign_id = r.campaign_id  
order by roi_pct desc;
```

	campaign_name	total_ad_spend	total_ad_revenue	roi_pct
▶	Autumn Fashion Launch 2024	228931.82	902677.91	294.30
	Mothers Day Gifting 2024	291734.08	1074774.17	268.41
	Boxing Day & New Year Sale 2025	536373.24	1913602.49	256.77
	Boxing Day & New Year Sale 2024	470126.04	1675771.71	256.45
	Mothers Day Gifting 2022	273936.10	932594.17	240.44
	EOFY Stocktake Sale 2024	680547.50	2279203.19	234.91
	Fathers Day Gifting 2023	254239.23	851076.10	234.75
	Mothers Day Gifting 2026	356668.76	1141349.93	220.00
	Autumn Fashion Launch 2023	223493.25	715110.07	219.97
	Spring Racing & Click Frenzy 2024	521827.94	1611227.29	208.77
	Autumn Fashion Launch 2026	271347.96	835184.78	207.79
	Fathers Day Gifting 2024	262419.32	804466.16	206.56
	Spring Racing & Click Frenzy 2023	471606.22	1417022.50	200.47
	Boxing Day & New Year Sale 2021	390430.68	1169640.44	199.58
	Autumn Fashion Launch 2021	183112.06	536508.06	192.99

Justification. ROI relates each campaign’s platform-reported ad revenue to its allocated budget, ranking campaigns on efficiency rather than raw size. This lets spend be steered toward the formats that return the most per dollar, which is the core question ahead of EOFY.

Insight. Every campaign is profitable, with ROI above 190%. Autumn Fashion Launch 2024 is the standout at 294.30%, followed by Mothers Day Gifting 2024 (268.41%) and Boxing Day

& New Year Sale 2025 (256.77%). The most recent EOFY benchmark, EOFY Stocktake Sale 2024, returned 234.91% - solid but mid-pack, signalling room to lift EOFY efficiency.

Addresses the concern? Yes. It ranks all campaigns by profitability and isolates the most efficient formats to learn from.

1.8 Business concern 3

Name and Description: Revenues of converted orders. We want to know the number of orders on the website that are ad-influenced (customers clicking on advertisements).

```
• with attributed as (  
    select distinct c.campaign_id, c.campaign_name, ao.order_id  
    from campaigns c  
    join advertisements a on c.campaign_id = a.campaign_id  
    join advertisement_order ao on a.ad_id = ao.ad_id  
),  
order_rev as (  
    select op.order_id, sum(op.quantity * op.gross_price) as order_revenue  
    from order_products op  
    join orders o on op.order_id = o.order_id  
    where o.order_status = 'Completed'  
    group by op.order_id  
)  
select  
    at.campaign_name,  
    count(distinct at.order_id) as total_attributed_orders,  
    round(sum(orv.order_revenue), 2) as total_actual_sales_revenue  
from attributed at  
join order_rev orv on at.order_id = orv.order_id  
group by at.campaign_id, at.campaign_name  
order by total_actual_sales_revenue desc;
```

	campaign_name	total_attributed_orders	total_actual_sales_revenue
▶	Mothers Day Gifting 2026	276	206857.83
	Autumn Fashion Launch 2026	238	187865.00
	EOFY Stocktake Sale 2025	201	145020.90
	Spring Racing & Click Frenzy 2025	201	133868.48
	EOFY Stocktake Sale 2024	156	129336.92
	Boxing Day & New Year Sale 2025	133	106112.71
	Autumn Fashion Launch 2025	133	97768.50
	Mothers Day Gifting 2025	104	83247.28
	Boxing Day & New Year Sale 2024	103	82084.49
	Spring Racing & Click Frenzy 2024	107	77336.30
	Fathers Day Gifting 2025	92	70174.67
	Mothers Day Gifting 2024	87	68752.79
	Spring Racing & Click Frenzy 2023	87	64752.65
	Fathers Day Gifting 2024	69	58839.38
	EOFY Stocktake Sale 2023	82	57360.62

Justification. Business concern 2 relies on platform-reported revenue, which can overstate impact. This query instead ties ad clicks to actual completed orders in the transactional tables, measuring the real downstream sales advertising produced and how many orders it influenced.

Insight. Mothers Day Gifting 2026 generated the most ad-attributed sales (\$206,858 from 276 orders), ahead of Autumn Fashion Launch 2026 (\$187,865). For EOFY specifically, EOFY Stocktake Sale 2025 ranks 3rd (\$145,021 from 201 orders) and EOFY 2024 5th (\$129,337 from 156 orders), confirming that EOFY advertising reliably converts into genuine revenue.

Addresses the concern? Yes. It quantifies both the volume and the dollar value of ad-influenced orders, grounded in real transactions rather than platform estimates.

1.9 Business concern 4

Name and Description: Customer Lifetime Value (LTV) by Loyalty Tier. This query identifies which membership tier generates the most value per customer to help the marketing team prioritize retention efforts.

```
with cust_rev as (  
  select c.customer_id, c.loyalty_tier,  
         coalesce(sum(op.quantity * op.gross_price), 0) as revenue  
  from customers c  
  left join orders o on c.customer_id = o.customer_id and o.order_status = 'Completed'  
  left join order_products op on o.order_id = op.order_id  
  group by c.customer_id, c.loyalty_tier  
)  
select loyalty_tier,  
       count(*) as customer_count,  
       round(avg(revenue), 2) as avg_ltv  
from cust_rev  
group by loyalty_tier  
order by avg_ltv desc;
```

	loyalty_tier	customer_count	avg_ltv
▶	Gold	57	46369.37
	Silver	99	19268.31
	Bronze	244	8731.76

Justification. Retention budgets are finite, so the team must know which loyalty tier returns the most value per customer before deciding where to concentrate retention spend.

Insight. Gold members are worth an average \$46,369 each - 2.4 times Silver (\$19,268) and 5.3 times Bronze (\$8,732) - yet there are only 57 of them against 244 Bronze. A small Gold cohort carries a disproportionate share of value.

Addresses the concern? Yes. It ranks tiers cleanly by per-customer value and flags Gold as the clear retention priority.

1.10 Business concern 5

Name and Description: Most Engaging Device Form. This query analyzes which devices (Desktop, Phone, or Tablet) customers use to interact with advertisements and the duration of those engagements.

```
select
    device,
    count(interaction_id) as total_interactions,
    round(avg(duration_seconds), 2) as avg_engagement_seconds
from interaction_events
group by device
order by avg_engagement_seconds desc;
```

	device	total_interactions	avg_engagement_seconds
►	Desktop	4846	61.35
	Tablet	1951	40.51
	Phone	8203	28.10

Justification. Knowing which device captures attention guides both creative format - richer desktop assets versus fast-loading mobile - and where interactive ad formats are best placed.

Insight. Phones generate by far the most interactions (8,203) but the shortest average engagement (28.1 seconds). Desktop is the opposite, with fewer interactions (4,846) but more than double the dwell time (61.4 seconds); tablet sits in between (40.5 seconds). Mobile delivers reach, desktop delivers depth.

Addresses the concern? Yes. It identifies the most engaging device while surfacing a reach-versus-depth trade-off that should shape creative strategy.

1.11 Business concern 6

Name and Description: Most Popular Payment Method. This query tracks the frequency and revenue contribution of different payment options used on the David Jones website.

```
select
  o.payment_method,
  count(distinct o.order_id) as total_orders,
  count(distinct o.customer_id) as total_customers_used,
  round(sum(op.quantity * op.gross_price), 2) as total_revenue,
  round(sum(op.quantity * op.gross_price) / nullif(count(distinct o.order_id), 0), 2) as avg_order_value
from orders o
join order_products op on o.order_id = op.order_id
where o.order_status = 'Completed'
group by o.payment_method
order by total_revenue desc;
```

	payment_method	total_orders	total_customers_used	total_revenue	avg_order_value
►	Debit/Credit Card	4390	392	3293511.90	750.23
	David Jones Credit Card	2578	348	2046933.33	794.00
	Gift Card	1155	365	851360.38	737.11
	Others	688	311	489360.93	711.28

Justification. Payment mix signals basket behaviour and the value of the store-card programme. Pairing usage with average order value shows not just the most common method but the method customers spend the most through.

Insight. Debit/Credit Card drives the most revenue (\$3.29M) and the most orders, but the David Jones Credit Card produces the highest average order value (\$794 versus \$750), implying store-card holders spend more per visit. Gift Card and Others trail on both volume and basket size.

Addresses the concern? Yes. It ranks methods by frequency and revenue and surfaces the store card as a distinctly high-basket segment worth promoting.

1.12 Business concern 7

Name and Description: Best seller category. This query identifies which product categories are the main drivers of revenue and unit volume for the online website.

```
select
  p.category,
  sum(op.quantity) as total_units_sold,
  round(sum(op.quantity * op.gross_price), 2) as total_revenue,
  round(sum(op.quantity * (op.gross_price - p.unit_cost)), 2) as gross_profit
from products p
join order_products op on p.product_id = op.product_id
join orders o on op.order_id = o.order_id
where o.order_status = 'Completed'
group by p.category
order by total_revenue desc;
```

	category	total_units_sold	total_revenue	gross_profit
►	Womenswear	5494	1738155.62	842593.04
	Beauty	5533	1328618.35	683260.13
	Menswear	2499	859410.77	426520.05
	Accessories	1665	761065.97	389397.00
	Electronics	826	695044.66	318426.50
	Footwear	2471	580406.14	277719.14
	Home	2091	513536.98	255920.97
	Food and Wine	575	144918.22	75398.29
	Kids	582	60009.83	30869.67

Justification. EOFY merchandising and ad creative should foreground the categories that actually drive revenue and margin, not simply those that move the most units.

Insight. Womenswear is the clear leader on both revenue (\$1.74M) and gross profit (\$843k). Beauty sells the most units (5,533) and ranks second on revenue. Menswear and Accessories follow, while Kids and Food & Wine contribute the least on every measure.

Addresses the concern? Yes. It pinpoints the revenue and profit drivers that EOFY should be built around.

1.13 Business concern 8

Name and Description: Platform and Device Synergy. This query examines the conversion efficiency of specific platform-device combinations to find the ‘sweet spot’ for ad placement.

```
select
  a.platform,
  ap.device,
  sum(ap.impressions) as total_impressions,
  sum(ap.conversions) as total_conversions,
  round(sum(ap.conversions) / nullif(sum(ap.impressions), 0) * 100, 2) as conversion_efficiency_pct
from advertisements a
join advertisement_performance ap on a.ad_id = ap.ad_id
group by a.platform, ap.device
order by conversion_efficiency_pct desc;
```

	platform	device	total_impressions	total_conversions	conversion_efficiency_pct
►	Instagram	Phone	62480512	568751	0.91
	Instagram	Tablet	61896910	558956	0.90
	Instagram	Desktop	58007525	519611	0.90
	WeChat	Desktop	46292308	361006	0.78
	WeChat	Tablet	46285719	353394	0.76
	WeChat	Phone	43038513	324101	0.75
	Facebook	Phone	55051784	408438	0.74
	Facebook	Tablet	53212566	388851	0.73
	Facebook	Desktop	51267616	375216	0.73
	X	Phone	44069873	87158	0.20
	X	Tablet	42245640	80000	0.19
	X	Desktop	47228341	89233	0.19

Justification. Aggregate platform metrics hide where conversion really happens. Crossing platform with device finds the specific placement that turns impressions into orders most efficiently - the true sweet spot for ad delivery.

Insight. Instagram dominates on every device, with Instagram-on-Phone the single best combination at 0.91% conversion efficiency, narrowly ahead of Instagram tablet and desktop (0.90%). WeChat and Facebook cluster in the mid-0.7s, while every X placement collapses to roughly 0.19-0.20%.

Addresses the concern? Yes. It isolates Instagram/Phone as the optimal placement and confirms X is weak across all devices.

1.14 Business concern 9

Name and Description: Highest Revenue by Locations. This query identifies which Australian cities contribute the most to the online revenue stream.

```
select
  c.location,
  count(distinct o.order_id) as total_orders,
  round(sum(op.quantity * op.gross_price), 2) as total_revenue,
  round(sum(op.quantity * op.gross_price) / nullif(count(distinct o.order_id), 0), 2) as avg_order_value
from customers c
join orders o on c.customer_id = o.customer_id
join order_products op on o.order_id = op.order_id
where o.order_status = 'Completed'
group by c.location
order by total_revenue desc;
```

	location	total_orders	total_revenue	avg_order_value
►	Sydney CBD NSW	1177	860485.81	731.08
	Melbourne CBD VIC	878	676291.41	770.26
	Chatswood NSW	664	489888.43	737.78
	Chadstone VIC	575	458981.41	798.23
	Brisbane CBD QLD	597	456965.57	765.44
	Parramatta NSW	550	427764.93	777.75
	Perth CBD WA	496	390748.62	787.80
	Gold Coast QLD	451	370104.22	820.63
	Canberra ACT	367	300531.37	818.89
	Highpoint VIC	374	300432.92	803.30
	Bondi Junction NSW	392	291378.85	743.31
	Newcastle NSW	339	261251.14	770.65
	Doncaster VIC	320	232466.52	726.46
	Macquarie NSW	297	226959.97	764.17
	Adelaide CBD SA	293	219549.33	749.32

Justification. Geographic revenue concentration informs where to weight regional ad spend and where premium messaging is most likely to land on higher-value baskets.

Insight. Sydney CBD is the top market (\$860k from 1,177 orders), ahead of Melbourne CBD (\$676k). High-volume metros run lower average baskets (Sydney \$731), while smaller affluent areas convert into larger ones - Gold Coast (\$821) and Canberra (\$819) lead on average order value.

Addresses the concern? Yes. It ranks cities by revenue and reveals a useful volume-versus-basket pattern across regions.

1.15 Business concern 10

Name and Description: Cost-per-Acquisition (CPA) by platform. This query measures the average cost of advertising to achieve an actual order.

```

with spend as (
  select platform, sum(allocated_budget) as total_spend
  from advertisements
  group by platform
),
conv as (
  select a.platform, sum(ap.conversions) as total_conversions
  from advertisements a
  join advertisement_performance ap on a.ad_id = ap.ad_id
  group by a.platform
)
select s.platform, s.total_spend, c.total_conversions, round(s.total_spend / nullif(c.total_conversions, 0), 2) as cpa
from spend s
join conv c on s.platform = c.platform
order by cpa asc;

```

	platform	total_spend	total_conversions	cpa
▶	Instagram	3276625.56	1647318	1.99
	Facebook	3034918.75	1172505	2.59
	WeChat	2823354.34	1038501	2.72
	X	3053757.23	256391	11.91

Justification. CPA reduces spend and conversions to the cost of winning a single customer, the cleanest efficiency metric for allocating a finite EOFY budget across platforms.

Insight. Instagram acquires customers most cheaply at \$1.99, ahead of Facebook (\$2.59) and WeChat (\$2.72). X is a clear outlier at \$11.91 - roughly six times Instagram - despite a comparable spend level.

Addresses the concern? Yes. It quantifies acquisition cost per platform and flags X as the inefficient channel to wind back.

1.16 Business concern 11

Name and Description: Ads ranking by CTR and budget. This query ranks the effectiveness of each advertisement in regarding of their CTR and budget.

```
with ad_perf as (
  select a.ad_id, a.platform, a.allocated_budget,
         sum(ap.impressions) as impressions,
         sum(ap.clicks) as clicks,
         round(sum(ap.clicks) / nullif(sum(ap.impressions), 0) * 100, 2) as ctr_pct
  from advertisements a
  join advertisement_performance ap on a.ad_id = ap.ad_id
  group by a.ad_id, a.platform, a.allocated_budget
)
select ad_id, platform, allocated_budget, impressions, clicks, ctr_pct,
       rank() over (order by ctr_pct desc) as ctr_rank,
       rank() over (order by allocated_budget desc) as budget_rank
from ad_perf
order by ctr_rank;
```

	ad_id	platform	allocated_budget	impressions	clicks	ctr_pct	ctr_rank	budget_rank
►	8	Instagram	35371.88	1783961	115552	6.48	1	147
	164	Instagram	70018.37	4165753	269185	6.46	2	80
	112	Instagram	65309.79	3938453	254590	6.46	2	88
	99	Instagram	120210.46	8418861	542208	6.44	4	14
	103	Instagram	36451.27	2267361	145487	6.42	5	144
	134	Instagram	147191.61	8557990	546918	6.39	6	3
	145	Instagram	111502.88	5462759	348413	6.38	7	22
	142	Instagram	56492.11	3115121	196595	6.31	8	104
	43	Instagram	57792.32	2923470	183175	6.27	9	102
	77	Instagram	41494.07	2439678	152707	6.26	10	134
	167	Instagram	79569.38	3922302	243044	6.20	11	69
	41	Instagram	32928.48	1945576	120549	6.20	11	150
	159	Instagram	101851.18	5413357	334740	6.18	13	35
	131	Instagram	74094.11	4895830	302477	6.18	13	74
	156	Instagram	61932.25	3402000	209934	6.17	15	97

Justification. Comparing each ad's CTR rank against its budget rank exposes misallocation: high-performing creatives that are underfunded, or weaker ads absorbing disproportionate budget.

Insight. The highest-CTR ads are all on Instagram, but funding does not follow performance. The single best ad by CTR (ad 8 at 6.48%) ranks only 147th on budget, while well-aligned performers such as ad 134 (CTR rank 6, budget rank 3) are properly funded. Several top creatives are starved of spend.

Addresses the concern? Yes. It ranks ads by effectiveness and, more usefully, reveals an actionable mismatch between ad quality and budget.

1.17 Business concern 12

Name and Description: Most and least profitable product by month. This query ranks products' profitability by month, showing which item generates the most and least profit in a month and year.

```
with prod_year as (  
    select p.product_id, p.product_name,  
           date_format(o.order_date, '%Y') as yr,  
           sum(op.quantity * op.gross_price) as gross_revenue,  
           sum(op.quantity * p.unit_cost) as total_cost  
    from order_products op  
    join orders o on op.order_id = o.order_id  
    join products p on op.product_id = p.product_id  
    where o.order_status = 'Completed'  
    group by p.product_id, p.product_name, yr  
)  
,  
line_disc as (  
    select opd.product_id,  
           date_format(o.order_date, '%Y') as yr,  
           sum(opd.product_discount_amount) as line_discount  
    from order_product_discount opd  
    join orders o on opd.order_id = o.order_id  
    group by opd.product_id, yr  
)
```

```

prod_profit as (
    select py.yr, py.product_name,
           py.gross_revenue - coalesce(ld.line_discount, 0) as net_revenue,
           py.gross_revenue - coalesce(ld.line_discount, 0) - py.total_cost as profit
    from prod_year py
    left join line_disc ld on py.product_id = ld.product_id and py.yr = ld.yr
),
ranked as (
    select yr, product_name,
           round(net_revenue, 2) as net_revenue,
           round(profit, 2) as profit,
           rank() over (partition by yr order by profit desc) as most_profitable_rank,
           rank() over (partition by yr order by profit asc) as least_profitable_rank
    from prod_profit
)
select yr, product_name, net_revenue, profit,
       most_profitable_rank, least_profitable_rank,
       case when most_profitable_rank <= 3 then 'Top 3'
            else 'Bottom 3' end as profit_segment
from ranked
where most_profitable_rank <= 3
     or least_profitable_rank <= 3
order by yr, most_profitable_rank;

```


	yr	product_name	net_revenue	profit	most_profitable_rank	least_profitable_rank	profit_segment
►	2021	Dyson Electronics Item 72	7927.44	3506.80	1	91	Top 3
	2021	Coach Accessories Item 83	6910.08	3185.66	2	90	Top 3
	2021	La Mer Beauty Item 3	5681.35	2646.07	3	89	Top 3
	2021	Sol Sana Footwear Item 50	224.93	76.56	89	3	Bottom 3
	2021	Maison Wine Food and Wine Item 98	154.10	70.02	90	2	Bottom 3
	2021	Penfolds Food and Wine Item 100	54.77	25.90	91	1	Bottom 3
	2022	Dyson Electronics Item 78	27489.11	11484.91	1	100	Top 3
	2022	Sonos Electronics Item 73	11716.47	6039.87	2	99	Top 3
	2022	La Mer Beauty Item 14	12732.31	6000.91	3	98	Top 3
	2022	Seed Heritage Kids Item 89	243.20	131.65	98	3	Bottom 3
	2022	Penfolds Food and Wine Item 100	319.69	117.60	99	2	Bottom 3
	2022	Seed Heritage Kids Item 93	117.57	37.86	100	1	Bottom 3
	yr	product_name	net_revenue	profit	most_profitable_rank	least_profitable_rank	profit_segment
	2023	Camilla Womenswear Item 35	26773.76	13217.93	1	100	Top 3
	2023	Coach Accessories Item 83	25249.87	10884.25	2	99	Top 3
	2023	La Mer Beauty Item 6	16992.94	10237.54	3	98	Top 3
	2023	Polo Kids Kids Item 94	1227.10	538.23	98	3	Bottom 3
	2023	Seed Heritage Kids Item 93	678.99	253.87	99	2	Bottom 3
	2023	Penfolds Food and Wine Item 100	354.54	123.58	100	1	Bottom 3
	2024	La Mer Beauty Item 14	51931.78	23996.47	1	100	Top 3
	2024	Camilla Womenswear Item 22	40188.47	20596.55	2	99	Top 3
	2024	Dyson Electronics Item 72	42042.65	20570.97	3	98	Top 3
	2024	Seed Heritage Kids Item 89	1173.45	615.70	98	3	Bottom 3
	2024	Seed Heritage Kids Item 93	1037.61	453.07	99	2	Bottom 3
	2024	Penfolds Food and Wine Item 100	1076.39	383.51	100	1	Bottom 3
	2025	Dyson Electronics Item 72	60968.51	30024.03	1	100	Top 3
	2025	Sass and Bide Womenswear Item 27	48169.05	29034.24	2	99	Top 3
	2025	Camilla Womenswear Item 35	60316.40	28575.92	3	98	Top 3
	2025	Seed Heritage Kids Item 89	1503.69	812.08	98	3	Bottom 3
	2025	Penfolds Food and Wine Item 100	1429.61	505.77	99	2	Bottom 3
	2025	Seed Heritage Kids Item 93	809.76	331.50	100	1	Bottom 3
	2026	Camilla Womenswear Item 35	48816.97	23358.46	1	100	Top 3
	2026	Dyson Electronics Item 72	43725.29	22253.61	2	99	Top 3
	2026	La Mer Beauty Item 10	51728.10	21413.22	3	98	Top 3
	2026	Seed Heritage Kids Item 89	1108.42	617.60	98	3	Bottom 3
	2026	Seed Heritage Kids Item 93	1133.27	469.02	99	2	Bottom 3
	2026	Penfolds Food and Wine Item 100	655.26	251.08	100	1	Bottom 3

Justification. Partitioning profit rankings by year shows which products consistently lead or lag over time, guiding which lines to feature in EOFY and which to rationalise or bundle.

Insight. A stable set of winners recurs every year - Dyson (Electronics), La Mer (Beauty), Camilla and Sass & Bide (Womenswear), and Coach (Accessories). The persistent laggards are Seed Heritage and Polo (Kids) and Penfolds (Food & Wine). In 2026 the top three are Camilla Womenswear (\$23,358 profit), Dyson Electronics (\$22,254) and La Mer Beauty (\$21,413).

Addresses the concern? Yes. It identifies hero products to push and chronic underperformers to bundle or discontinue.

1.18 Business concern 13

Name and Description: Customer tenure and Loyalty tier. Which loyalty tier has the most customers and which tier stays with the business the longest number of days.

```
select
  c.loyalty_tier,
  count(*) as customers,
  round(avg(datediff(curdate(), c.signup_date)), 0) as avg_days_with_brand,
  min(datediff(curdate(), c.signup_date)) as min_days,
  max(datediff(curdate(), c.signup_date)) as max_days
from customers c
group by c.loyalty_tier
order by avg_days_with_brand desc;
```

	loyalty_tier	customers	avg_days_with_brand	min_days	max_days
►	Bronze	244	1156	27	1999
	Silver	99	1108	40	2000
	Gold	57	1090	36	1976

Justification. This tests whether loyalty tier is earned through longevity or through spend, which determines whether tier value is driven by retention or by acquisition.

Insight. Bronze holds the most customers (244) and, unexpectedly, the longest average tenure (1,156 days), marginally ahead of Silver (1,108) and Gold (1,090). Tenure is essentially flat across tiers, meaning tier is driven by spending intensity, not time with the brand.

Addresses the concern? Yes, for the counts and tenure question, and it adds the important insight that longevity does not explain tier status.

1.19 Business concern 14

Name and Description: All-time average daily profit by customer. This query shows who is David Jones's highest spender. Will customers that stay with the brand longer spend more than newer ones?

	customer_id	customer_name	loyalty_tier	days_with_brand	total_profit	avg_daily_profit
►	123	Mohammed Le	Gold	66	32886.57	498.28
	338	Wei Adams	Gold	36	17403.29	483.42
	178	Jack Young	Gold	52	23548.49	452.86
	342	Hung Brown	Gold	61	26976.09	442.23
	138	Zoe Anderson	Gold	80	25736.72	321.71
	167	Oliver Green	Gold	103	24443.12	237.31
	368	Daniel Carter	Gold	120	23737.60	197.81
	271	Emily King	Silver	40	7250.02	181.25
	181	Diego Tran	Silver	43	7510.01	174.65
	88	Amelia Nelson	Bronze	36	6086.45	169.07
	131	Hung Le	Bronze	34	5312.75	156.26
	175	Khoa Green	Silver	71	10256.69	144.46
	227	Emily Hall	Silver	57	7424.66	130.26
	362	Max Nelson	Gold	205	23394.72	114.12
	24	Max Baker	Silver	98	10631.80	108.49
	customer_id	customer_name	loyalty_tier	days_with_brand	total_profit	avg_daily_profit
	282	Khoa King	Bronze	27	2836.79	105.07
	266	Ruby Taylor	Gold	332	28520.64	85.91
	240	Liam Walker	Bronze	57	4286.44	75.20
	400	William Baker	Silver	154	11139.13	72.33
	109	Sofia Thompson	Silver	146	9104.02	62.36
	236	Harper Allen	Bronze	69	3721.50	53.93
	158	Harper Anderson	Bronze	81	4277.43	52.81
	113	Leo Young	Silver	186	9053.14	48.67
	118	Isla Lee	Silver	235	11167.86	47.52
	12	Hannah Nguyen	Gold	539	24632.54	45.70
	226	Bao Wilson	Bronze	107	4699.13	43.92
	81	Zoe Lewis	Gold	493	21159.85	42.92
	41	Olivia Nguyen	Gold	672	27959.50	41.61
	179	Aarav King	Gold	635	26223.86	41.30
	352	Oliver Khan	Gold	684	27580.38	40.32

	customer_id	customer_name	loyalty_tier	days_with_brand	total_profit	avg_daily_profit
	18	Hiroshi Wright	Gold	807	31936.07	39.57
	375	Grace Anderson	Silver	182	6904.20	37.94
	94	Lucas Williams	Gold	575	21533.27	37.45
	206	Isla Lee	Bronze	187	6989.15	37.38
	77	Hannah Garcia	Silver	292	10138.98	34.72
	267	Ella Adams	Gold	839	28429.69	33.89
	339	Wei Tran	Gold	899	30120.28	33.50
	30	Max Adams	Gold	921	29926.13	32.49
	86	Hiroshi Baker	Gold	1016	30194.70	29.72
	56	Charlotte Tho...	Bronze	180	5333.67	29.63
	149	Mai Hall	Gold	842	24933.04	29.61
	211	Hannah King	Gold	708	19449.68	27.47
	335	Hannah Le	Silver	449	12126.01	27.01
	241	Jack Scott	Bronze	204	5496.62	26.94
	395	Scarlett Walker	Gold	1127	30281.09	26.87

Justification. Average daily profit normalises each customer's spend by their tenure, revealing spending velocity and testing whether longer-tenured customers are genuinely worth more on a day-to-day basis.

Insight. The highest-velocity customers are recent Gold members - for example one Gold customer generated \$498 per day across just 66 days - while several long-tenured customers show far lower daily profit (one 1,127-day customer at \$27 per day). Longer tenure does not translate into higher daily spend; the most valuable customers are quickly-acquired, high-spending Gold members.

Addresses the concern? Yes. It answers who the highest spenders are and overturns the assumption that tenure predicts value, reframing it around acquiring high-velocity spenders.

Recommendation

The 14 queries point to a consistent story: David Jones can significantly improve EOFY 2026 by reallocating budget toward channels and segments that already demonstrate superior efficiency, and away from those that drain spend. The following recommendations translate the findings into action.

1. **Reallocate budget from X to Instagram.** X is the weakest channel on every metric - CTR of 2.10% (Concern 1), conversion efficiency near 0.19% (Concern 8) and a CPA of \$11.91, roughly six times Instagram's \$1.99 (Concern 10). Redirecting X's spend toward Instagram, the leader on CTR, efficiency and acquisition cost, should lift conversions at a materially lower cost per customer.
2. **Concentrate Instagram spend on Phone and fund the proven creatives.** Instagram-on-Phone is the single best converting placement (Concern 8), yet several of the highest-CTR Instagram ads are underfunded, such as the top creative by CTR ranks only 147th by budget (Concern 11). Shifting budget into these starved high-CTR ads compounds the gains from the platform reallocation above.
3. **Pair Instagram for reach with WeChat for closing.** Instagram wins the click with a 6.02% CTR, but WeChat converts clicks into orders best at 24.98% (Concern 1). A sequenced funnel - Instagram for awareness and WeChat retargeting to close - plays each channel to its proven strength rather than treating them as interchangeable.
4. **Build EOFY around Gold members and the store card.** Gold customers are worth an average \$46,369 each (Concern 4), and the fastest daily spenders are recent Gold acquisitions (Concern 14). Offer Gold members early EOFY access and VIP previews, and promote David Jones Credit Card sign-ups at checkout, since store-card holders carry the highest average order value at \$794 (Concern 6).
5. **Merchandise proven categories and hero brands.** Lead EOFY creative with Womenswear and Beauty, the top performers on revenue, profit and units (Concern 7), and feature the consistent profit leaders Camilla, Sass & Bide, La Mer, Dyson and Coach (Concern 12). Bundle chronic laggards such as Kids and Food & Wine rather than giving them standalone spotlight.
6. **Tailor regional and device messaging.** Weight spends toward Sydney and Melbourne for volume, but run premium, higher-basket creative to Gold Coast, Canberra and Chadstone, which produce the largest average orders (Concern 9). Use richer desktop creative for depth and fast mobile formats for reach, reflecting the device engagement split (Concern 5).
7. **Set EOFY 2026 targets above the recent benchmark.** EOFY Stocktake Sale 2024 returned 234.91% ROI (Concern 2) and EOFY 2025 drove around \$145,000 in ad-attributed sales (Concern 3). With the reallocations above, EOFY 2026 should be targeted to exceed both the ROI and attributed-revenue baselines.